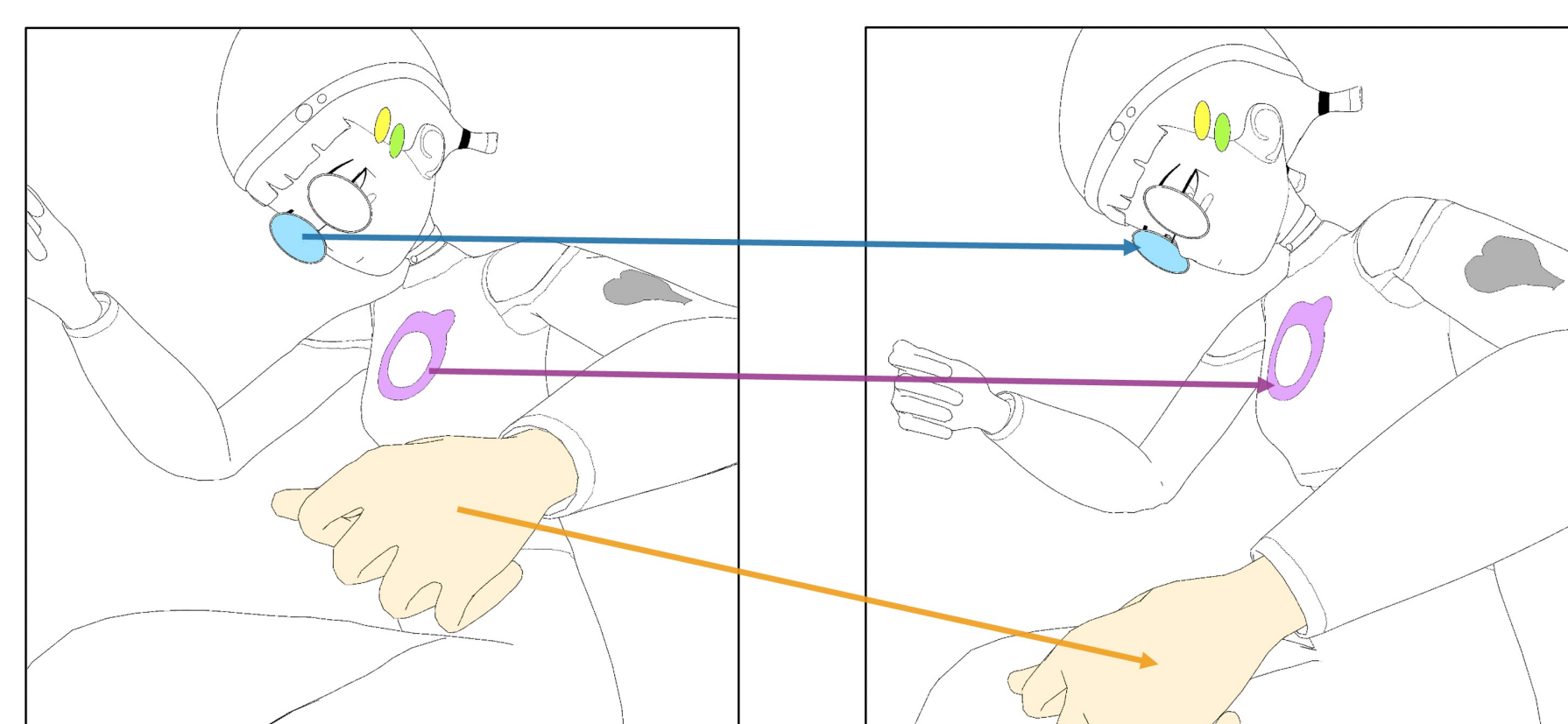


1. Introduction

Structural understanding of manga line art



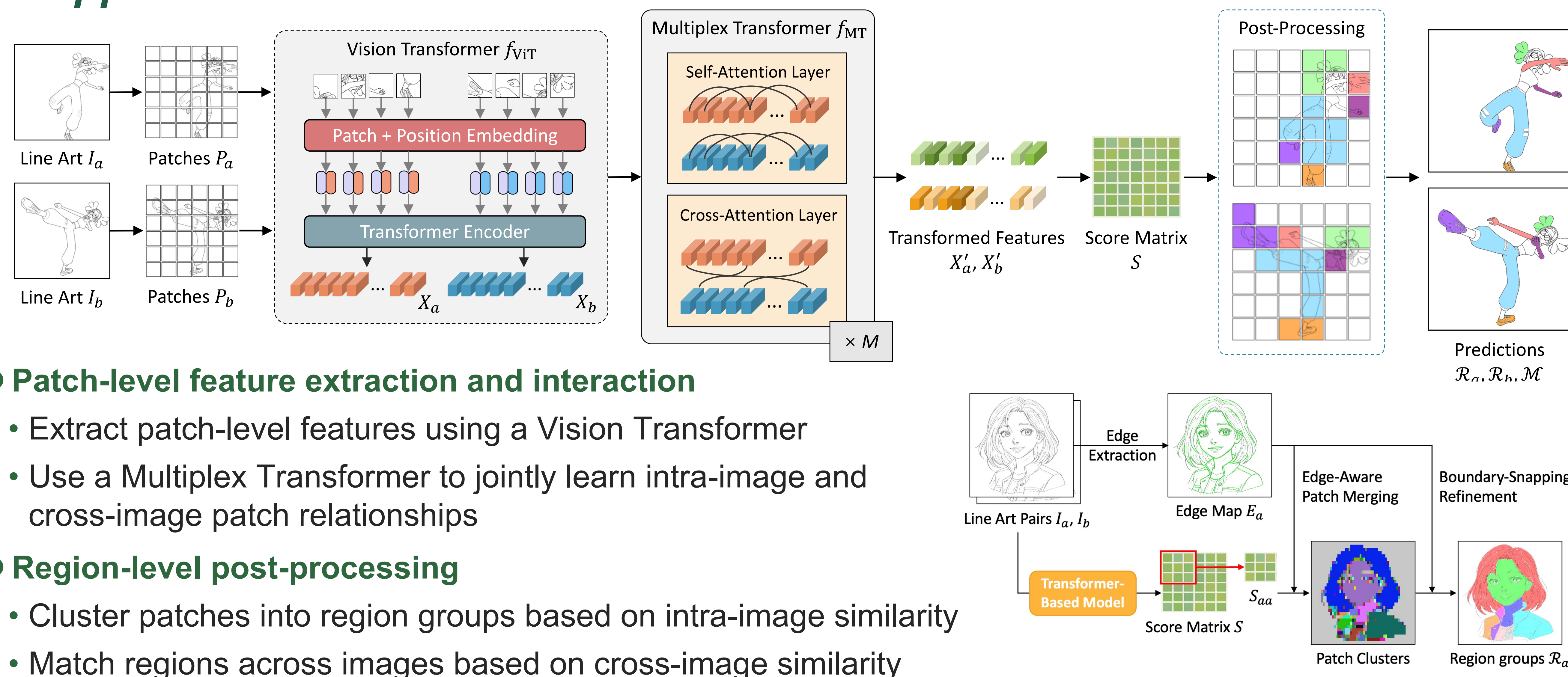
Task

- Input: A pair of manga line art images
- Segment each image into meaningful regions
- Match the corresponding regions

Application

- Line art colorization
- In-between frame generation
- Consistent editing across frames

3. Approach



Patch-level feature extraction and interaction

- Extract patch-level features using a Vision Transformer
- Use a Multiplex Transformer to jointly learn intra-image and cross-image patch relationships

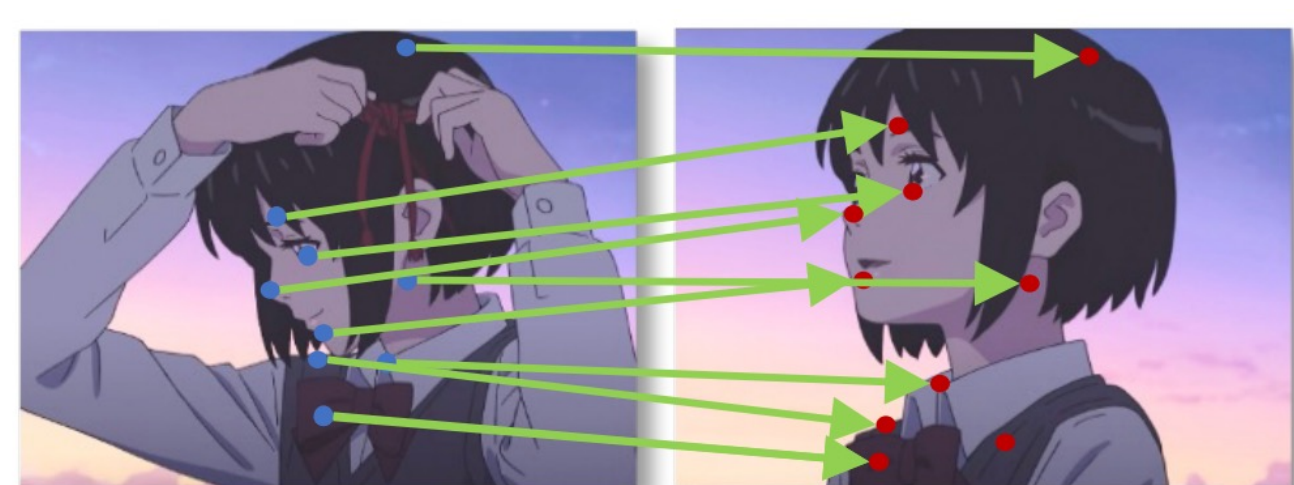
Region-level post-processing

- Cluster patches into region groups based on intra-image similarity
- Match regions across images based on cross-image similarity

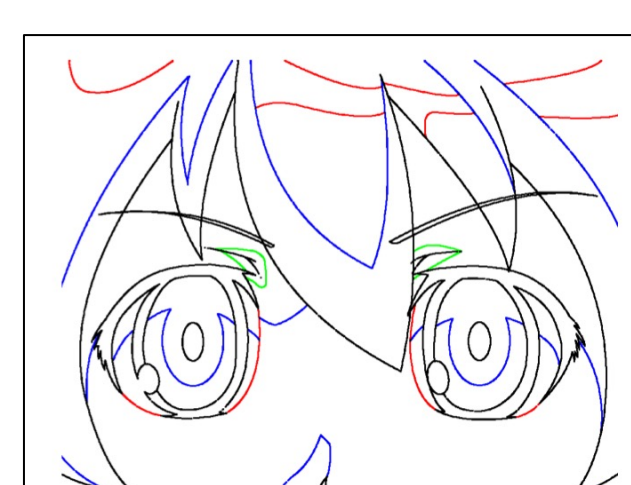
2. Motivation & Proposal

Limitations of existing methods

- Point-to-point matching^[1]



- Region-wise matching^[2]



Our works

- Propose a new task of predicting region-wise correspondence across manga line art
- Present a Transformer-based approach that achieves high accuracy
- Develop an automatic annotation pipeline and construct a manually refined benchmark dataset

4. Experiments

Datasets

- **3D-rendered line art:** PaintBucket-Character (PBC) dataset^[2]
- **Hand-drawn manga style line art:** ATD^[3], GenAI (synthetic dataset)

Region-level evaluation

- **Baseline:** Adapt a colorization-based method^[2] as a baseline
- **Intra-image region grouping via patch merging:** Adjusted Rand Index (ARI), mean IoU (mIoU), Cluster Ratio (CR)
- **Cross-image region correspondence prediction:** Region Accuracy

Dataset	Method	Matching Type	ARI	mIoU (P→G)	mIoU (G→P)	CR	Region Accuracy
GenAI	Baseline	Intra-img (I_a)	33.90	5.89	44.84	13.84	-
		Intra-img (I_b)	34.89	5.68	45.01	14.22	-
		Cross-img	-	-	-	-	72.36
	Ours	Intra-img (I_a)	46.23	32.50	40.09	1.70	-
		Intra-img (I_b)	50.92	32.69	40.90	1.66	-
		Cross-img	-	-	-	-	78.43

References

- [1] MangaNinja: Line Art Colorization with Precise Reference Following. Liu et al., CVPR2025.
 [2] Learning inclusion matching for animation paint bucket colorization. Dai et al., CVPR2024.
 [3] Deep Animation Video Interpolation in the Wild. Li et al., CVPR2021.



Project page

